brake force and lateral force get absorbed. Due to the shortened length of the upper arm the tire wear gets reduced.

Special Features

1. Since length of upper wishbone is shorter than the lower wishbone, wheels are slightly inclined. This leads to no slip in the tyre and the wheel track does not change and hence the tyre wear is reduced.
2. The two wishbone arms help in positioning the front wheels and they withstand the braking torque while braking.
3. There is no effect on one wheel while other wheel travels through ups and downs in the road.

7.3.2.2 Vertical guide suspension system

The simple layout is shown in figure 7.3.2.2. It is the first developed independent front suspension system. In this suspension the kingpin is directly attached to the cross member of the frame and spacing rod. The coil spring is fitted as shown in the figure.

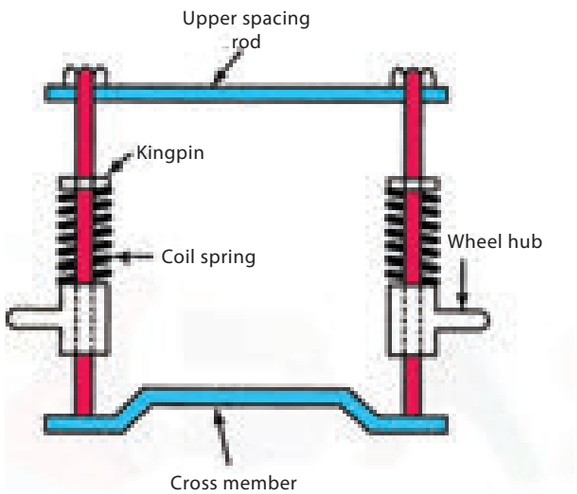


Figure 7.3.2.2 Vertical guide suspension system

When the wheel travel over the bumpy road, spring compress or expands and this makes the stub axle to slide up and down. In this type, caster angle, camber angle and wheel direction are remained unchanged as the wheel moves up and down. However, the disadvantage is less strength.

Notes

1. While the vehicle running, the sliding member is also moving up and down. Therefore, the spring gets compressed and elongated continuously.
2. When the specific wheel going through up and down, the sliding member is also moving up and down. However, the castor and camber angle will not change.
3. This is system cannot be used because of its less strength.
4. The efficiency of the system is based on the sliding member movement accuracy.
5. It is difficult to avoid the shock and less holding that can occur.

7.3.2.3 Trailing link types

Construction

The figure 7.3.2.3 shows the construction of the trailing link type. In this type of suspension system, coil springs are positioned between cross member and the trailing link. On the other side of the trailing link, wheels are fixed. In some vehicles, torsion bars are used instead of horizontal coil spring.

Working principle

When the vehicle wheel passing over the bumps and pits, trailing arm oscillates

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